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AUSTRALIAN LONGITUDES by “WIRE and WIRELESS”

Introduction

For around two thousand years, maps have contained lines indicating latitude and longitude that allow both the map maker and the map user to uniquely specify a position using this reference framework. In his circa 200AD work, known today as *Geography* and also known as *Geographia, Cosmographia, or Geographike Hyphegesis*, Claudius Ptolemy listed alphabetically the names of some 8000 places across the then known world along with their related latitude and longitude coordinates in degrees.

The work of early astronomers had established the Equator as the place where the sun was directly overhead twice each year and Ptolemy set this as the zero-degree parallel. He was free, however, to choose any meridian for the zero-degree longitude or *prime meridian*. Over time, the location of the prime meridian was moved from the Canary (Fortunate) Islands (Ptolemy’s choice), the Azores, Copenhagen, Jerusalem, St Petersburg, Pisa, Paris and Philadelphia to Greenwich in 1884. In that year at the International Meridian Conference in Washington, 22 of the 25 nations that attended agreed on Greenwich as the location of the prime meridian due to its popularity. France abstained from the vote and French maps continued to use the Paris meridian for several more decades.

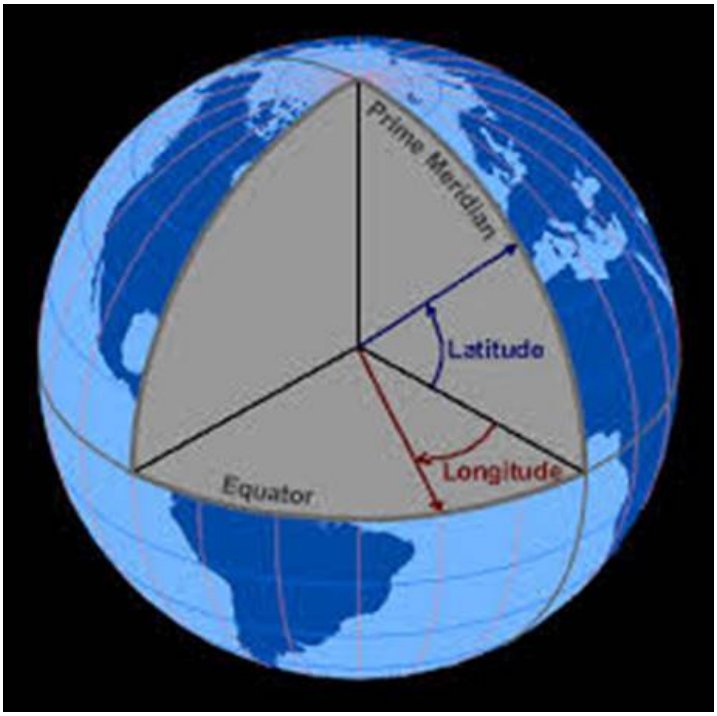


Figure 1 : Diagram showing the basic definition of Latitude and Longitude

Therein lies the fundamental difference between latitude and longitude. The zero degrees value of latitude is fixed by natural laws (orbit of the Earth) whereas the zero degrees value of longitude is fixed by international agreement.

Measuring latitude and longitude

A *noonsight* was the common way to find latitude. At local noon (the time that the sun was at its highest point in the sky at a given place on a given day), the angle between the sun and the horizon was measured, generally with a sextant. With a sextant the sun could be observed to rise, slow, stop then just before the sun started to descend was the instant of local noon. The angle read from the sextant at that instant gave the sun’s altitude. The sun’s declination was taken from an ordinary book of nautical tables and simple arithmetic gave latitude. The accuracy of the result was more reliant on the observer’s skill to judge local noon and read the angle whilst standing on a ship’s unstable deck, than on any of the mathematics.



Figure 2 : Using a Sextant for shipboard astronomical observation

Longitude, however, was a function of there being 24 hours in a day during which the Earth spins 360º (hence at 15º per hour). Thus, local noon is 1 hour later for every 15º of longitude east of Greenwich and conversely local noon is 1 hour earlier for every 15º of longitude west of Greenwich. Longitude's relationship and dependency on time was the reason that inaccurate longitude determination plagued mariners until transportable sea-faring clocks could be manufactured.

As maritime trade became more and more important and new territories were established, sooner or later an observatory was constructed near most main ports. With an array of instrumentation regular observations on the sun, moon and stars then commenced. Air pressure and temperature and even the local force of gravity were also amongst the regular readings recorded. With sufficient data an accurate latitude and longitude of the observatory and thus the port would be computed and refined over time with continued observations. These data were *fed* into maps, nautical almanacs, star catalogues etc to assist maritime navigation.

Between such ports, longitude could only be calculated from the ship's distance and direction travelled between local noon on successive days. Changing winds and currents affected the ship's speed and if cloud obscured the sun, local noon might not be observed for some days. Despite various devices being employed to attempt to obtain better information about the ship's daily progress, the ship's longitude was still only an estimate; albeit sometimes a very good estimate.

Mariner's clocks

John Harrison an English woodworker is credited with the manufacture of the first, portable clock able to keep relatively accurate time at sea. Harrison's knowledge of wood enabled him to use the strength and grain of Oak to carve the required gears with unbreakable teeth and self-lubrication. The English *Longitude Act* of 1714 offered a *reward* for a *Practical and Useful* means of determining longitude for which Harrison spent his whole adult life competing. Unfortunately his main rival, the fifth astronomer royal, the Reverend Dr Nevil Maskelyne, believed longitude could be solved with astronomy as was latitude, and Maskelyne's efforts to win the prize have been described by some as *foul play*. A suitably accurate astronomic solution for finding longitude at sea was never established and thus Harrison was finally rewarded for his clock developments in 1773.

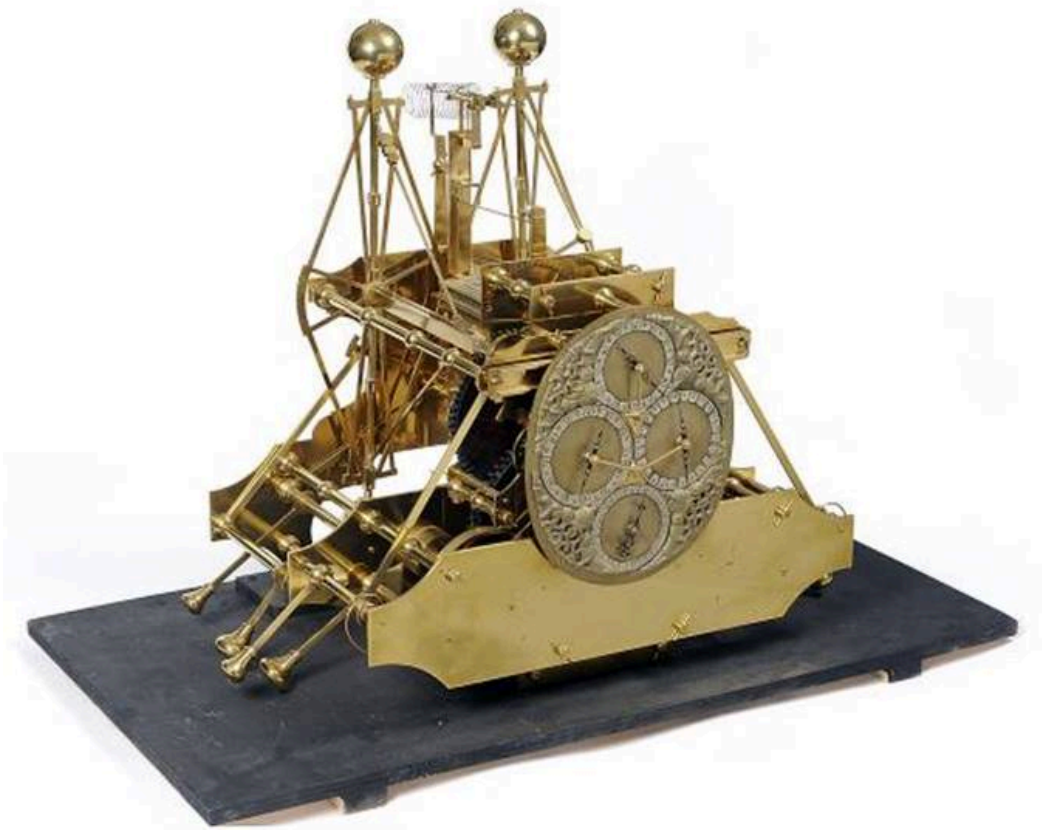


Figure 3 : A replica of John Harrison's first clock known as H1

In his efforts to develop a solution to longitude determination, Maskelyne left a valuable legacy that was used for nearly a century. The *Nautical Almanac and Astronomical Ephemeris* first published in 1767 and *inter alia* provided tables of lunar distances which gave the distance of the Moon from the Sun and selected stars suitable for lunar observations. Very simplistically, the position of the Moon changing relative to the *fixed* celestial background, was similar to the position of the hands of a clock relative to the dial, and could be used to determine time. Maskelyne's publication later became the standard almanac for mariners worldwide, and since it was based on the Royal Observatory, it helped lead to the international adoption of Greenwich Mean Time as the international standard.

The importance of Maskelyne's work was that the *lunar distance method or lunar distances* was a way of finding longitude without a clock. However, it was computationally intensive; prior to Maskelyne's tables 10 hours of computational effort per observation set was mentioned. With Maskeleyne's tables the calculation time was reduced to around 30 minutes. The *lunar distance method,* however, was only accurate to about 15 minutes of arc (25 kilometres) and during the period of *new Moon*, when the Moon is close to the Sun, no *lunar distance* observations could be performed. Grabowski *et al* (2009) gives more information on this complex astronomical observation.

Nevertheless, *lunar distances* were widely used at sea from 1767 until about 1850. Records also show that on the ill-fated 1860-61 Australian expedition of Robert O'Hara Burke and William John Wills, surveyor-navigator Wills used *lunar distances* to monitor the expedition's time and longitude and hence position and progress.

Time based longitude

While clocks successfully impacted marine navigation, precise longitude required very, very precise time. Recall that for every hour of time the Earth revolves through 15º of longitude. Thus in one second of time the change in longitude is 15 seconds of arc. One second of arc is approximately 30 metres so in one second of time the change in longitude is some 450 metres of distance. To know longitude to an accuracy of less than 5 metres, time needs to be known to less than 1/100th of a second (0.01s). Clocks keeping time to this level of accuracy did not appear until around the beginning of the 20th century.

(Note that it is usual to express latitude and longitude in degrees (and sub-units) of *arc*. Longitude however may also be expressed in hours (and sub-units) of *time*. In this paper the discussion will determine whether degrees or hours are used in the first instance with the alternative expression provided for completeness).

At the outset the *best* clocks could have errors that over an average ship's voyage meant that the ship's position was only known to within 20 kilometres. As clock manufacture improved, the ability to keep accurate time likewise improved but environmental factors like heat, cold, salt air, dust, and humidity still randomly affected time keeping performance.

At many ports today an old *time-ball* tower still exists (and may still operate) as a reminder of those days when the large ball on the tower dropped down its central support each afternoon at one o'clock to allow shipmasters to correct their clocks. At some ports a canon was fired as the signal and in some places both methods were used. As the article in Annexure A shows time-balls also allowed the general public to adjust their clocks and thereby bring uniformity to the recording of time in a region. The determination of the local time being 1PM was usually the responsibility of the local observatory. It must be

remembered, however, that these observatories acted in isolation as there was no independent means of verifying the accuracy of their work.



Figure 4 : Time-ball Tower, Williamstown, Melbourne

Between 1800 and 1850, reliable marine clocks started to emerge in quantity, such that it became possible to buy two or more relatively inexpensive clocks, that served as checks on each other, rather than purchasing an expensive sextant suitable for navigation by lunar distance. By 1850, however, the vast majority of maritime operators had ceased using the method of lunar distances. Mariners now set their clocks at their last port of call (generally to Greenwich Mean Time – GMT) via the time-ball signal, and on each day of their voyage at their local noon they noted the time on their clocks. As their clocks read GMT at local noon the difference in time gave them their longitude. For example, if it was local noon (1200hrs) but the ship's clock indicated it was 0200hrs GMT then they were 10 hours ahead of GMT. Hence the longitude was 150ºE (10hr x 15º per hr). If the ship's clock, however, had read 2200hrs GMT then they were 10 hours behind GMT and the longitude would have been 150ºW.



Figure 5 : Chronometers of various sizes and styles

The Electric Telegraph

The electric telegraph was a communication system that enabled a message to be transmitted via electric signals over wire from location to location (the signals were either voltaic or galvanic referring as to how the electric current was generated).

In 1830, American Joseph Henry, demonstrated the potential of the electromagnet invented in 1825 by British inventor William Sturgeon, for long distance communication. Henry sent an electronic current over 2 kilometres of wire to activate an electromagnet which caused a bell to strike. This same technology was also used by observatories to activate the time-balls. The observatory and the time-ball could now be some distance apart or the same signal could activate more than one time-ball.

Samuel Morse successfully exploited the electromagnet and bettered Joseph Henry's invention. Morse invented a telegraph system that was a practical and commercial success. Morse's system was first demonstrated in public in 1838 but it was another five years before the American Congress funded the construction of an experimental telegraph line from Washington to Baltimore, a distance of 65 kilometres. On May 1, 1844, this line was first used to dispatch news. The completed line was officially opened on May 24, 1844.

A couple of weeks later, in June 1844, Charles Wilkes used the telegraph line to establish the longitude of Baltimore. (Wilkes was a Commander in the US Navy and also explorer who in 1838 had led the US Exploring Expedition that discovered *inter alia* the continent of Antarctica).

Longitude by "Wire"

Wilkes adjusted a clock to local time, using astronomical observations, and installed it at the Washington telegraph office. Lieutenant Henry Eld, a veteran of Wilkes' expedition, did the same in Baltimore. For three days the officers at each end of the line took turns sending pulses every 10 seconds, in time with the *beat (tick)* of their clock. The officer on the receiving end tried to note the exact local time when he heard the electromagnet click. When the click came between the *beats* of his clock, he estimated the fraction of a second. When he finished, the officer sending the signals transmitted the exact local hour and seconds at which he had sent the 10 second sequences. The differences in time between the two stations were averaged to give the longitude. Wilkes reckoned that the Capitol in Washington was 1 minute 34.868 seconds of time west (0.39528 degrees west) of Battle Monument Square in Baltimore.

Unfortunately, the procedure was flawed in that it required human estimation of the fraction of the second. Experiments found that observer error could approach 0.2 seconds, far in excess of that which could be tolerated. The *star signal method* was first used in 1846. The difference in longitude between two stations, connected by telegraph, came from measuring the time difference between an agreed star(s) crossing the meridian (line of longitude) of the eastern station and then the meridian of the western station. A clock at one station only was used to record the instance of the star(s) crossing. The elegance of the *star signal method* was that errors in the star's position, as given in the star tables were irrelevant, as were any absolute errors of the clocks as only a single clock was employed. This clock, however, needed to be able to correctly measure the interval between meridian passages at the two stations.

To further remove human error from this methodology a *chronograph* with paper tape recorder was later used. The chronograph not only accurately maintained time but during observations at each *beat (tick)* of the chronograph a mark would be automatically made on a roll of paper tape. The observers' watching for the star(s) passage could also mark the tape at the instant of crossing via telegraph. Very accurate interpolation of the observers' marks against the seconds' marks on the paper tape gave the time of passage without the need for human estimation.





Figure 6 : Chronograph with paper tape recording (left) and (right) the Sydney Observatory Chronograph also with paper tape recording

Longitude by telegraph (wire) became an acceptable and accurate method of obtaining longitude as it allowed time differences to be measured with greater accuracy (averaged to less than 1/100th of as second) over any distance spanned by the telegraph.

Australian longitudes by “Wire”

The first astronomical observations known to be made on Australian soil were those by then Lieutenant James Cook in 1770. From these observations was derived the longitude of Fort Macquarie, Sydney as 151º 11’ 32” (10h 4m 46.14s) East of Greenwich. The Sydney Opera House now stands on this site.



Figure 7 : Fort Macquarie, Sydney, circa 1870

The *First Fleet*, commanded by Captain Phillip arrived at Port Jackson in 1788. After that Fleet's arrival a Colonel Collins reported that:… *"Among the buildings that were undertaken shortly after our arrival must be mentioned an observatory, which was marked out on the western point of the cove, to receive the astronomical instruments which had been sent out by the Board of Longitude, for the purpose of observing the comet which was expected to be seen about the end of this year (1788). The construction of this building was placed under the direction of Lieutenant Dawes, of the Marines who, having made this branch of science his peculiar study, was appointed by the Board of Longitude to make astronomical observations in this country."*

In 1793, the Spanish *Malaspina Expedition* spent two months at Sydney. Admiral Don Jose D'Espinosa on the *Corbetas Descubierta y Atrevida* recorded the longitude of Sydney. When this value was reduced to Fort Macquarie, it gave the value of 10h 4m 51.91s (151º 12' 59") which at the time was considered to be within a fraction of a second of time of the latest accepted value.

Thirty-three years after Cook, the longitude of Fort Macquarie, Sydney, was determined by Matthew Flinders as 151º 11' 49" (10h 4m 47.27s). The amount of Flinders' lunar observations was remarkable, both in quality and quantity. His value for the longitude was then considered to be within one mile (approximately 1 minute of arc of longitude or 1.5 kilometres) of the true value which, considering the instrumental limitations and the inaccuracy of the lunar tables in his day, *"may well be accepted as a result of the highest accuracy attainable at the time"*.

Fort Macquarie thus became the principal meridian in Australia. During Captain Fitzroy's famous second voyage in HMS *Beagle* in 1831-36 it was one of the points for which longitude was derived. The Beagle's second voyage determined the longitude of points around the globe by transportation of chronometers (a clock especially designed to be minimally affected by external factors and therefore able to maintain time more accurately). The longitude determined for Fort Macquarie was 10h 4m 32.14s (151º 08' 2.10").

In the latter half of the 1800s, government observatories were established as the Australian colonies were enacted. Until the year 1883 the adopted fundamental meridians of Australia were those of the Observatories of Sydney and Melbourne, and the longitude assigned to these meridians depended on the observation of *moon culminations* and *moon culminating stars* (*Culmination* is when the moon or star or both cross [transits] the meridian [line of longitude] of the observer).

The 141st meridian of east longitude was proclaimed as the eastern boundary of the Colony of South Australia in 1834. As discrepant positions were assigned to this meridian on different maps of the time, in 1839 Surveyor Charles James Tyers was commissioned by Sir George Gipps, Governor of New South Wales, to ascertain its actual location. Tyers adopted the longitude of Fort Macquarie and determined the longitude of a point on Batman's Hill, near Melbourne, by transportation of chronometers. He then used survey triangulation to fix the position of the 141st meridian, which he verified by sextant observations of lunar distances.

Unfortunately, the longitude value (151º 15' 14" or 10h 5m 0.93s) Tyers adopted for Fort Macquarie was later shown to be more than 4 kilometres in error. (This error was found in 1867 and further discussed below). Thus despite having carried out his own work with all the accuracy which was possible under the circumstances, the consequence of this error was that the boundary was fixed, and afterwards (in 1847) actually marked on the ground more than 3 kilometres to the west of the 141st meridian.

In April 1856 the Reverend William Scott took the position of colonial astronomer in New South Wales. On arrival with his family in Sydney he found the astronomical works were somewhat neglected. He supervised the building of the Sydney Observatory at Dawes Point, ensured the appointment of an Observatory Board and established the keeping of meteorological records throughout the colony. The Sydney Observatory still stands at Dawes Point today.

The first value for the longitude of the Sydney Observatory was obtained by Scott in 1858, and was derived from the observations of 21 transits of the moon. The result was 10h 4m 49.0s (151º 12’ 15”). In 1859 Scott further observed 38 moon culminations which gave as the resulting longitude of the Sydney Observatory 10h 4m 59.86s (151º 14’ 58”).

Edward James Stone was appointed chief assistant at the Royal Greenwich Observatory in 1860, and at once undertook the fundamental task of improving astronomical constants. In the light of his own work he *re-examined* Scott's Sydney 1859 and 1860 moon culminations and obtained a longitude for the Sydney Observatory of 10h 4m 47.32s (151º 11’ 49.8”) which was duly adopted.

The first determination of the difference in longitude between the Observatories of Sydney and Williamstown by the telegraphic exchange of clock signals, took place in 1861. The value obtained was Sydney-Williamstown -0h 24m 55.38s. This gave the longitude of Williamstown Observatory as 9h 39m 51.94s (144º 57’ 59”). (Williamstown is now a suburb of Melbourne and was the Victorian capital’s early port; the Williamstown time-ball tower still exists as shown in Figure 4).

The longitude of the Williamstown Observatory was, however, determined by observations of moon culminations in the years 1860, 1861, and 1862, resulting in the adopted value being 9h 39m 38.8s (144º 54’ 42”). Accurate triangulation enabled the difference in longitude between the Williamstown and Melbourne Observatories to be computed as +16.0s. When this difference was applied to the longitude of Williamstown it gave the longitude of the Melbourne Observatory as 9h 39m 54.8s (144º 58’ 42”) which was adopted until 1883.

The 1867 telegraphic exchange of clock signals between the Observatories of Melbourne and Adelaide resulted in a difference of -0h 25m 33.76s, which when applied to the longitude of Melbourne gave the longitude of Adelaide Observatory as 9h 14m 21s (138º 35’ 15”), which was also adopted until 1883.

The above astronomical work and the results of the Trigonometrical survey of Victoria, which connected the Melbourne Observatory with a western station of the survey near the South Australian boundary, indicated a problem with Tyers 1839 longitude value for Fort Macquarie. The boundary between the Colonies of Victoria and South Australia had been marked too far west (as explained above)!

The governments of the affected states (New South Wales, Victoria and South Australia) required the position of the 141st meridian to be re-determined and that their representatives George Roberts Smalley (Government Astronomer, NSW), Robert Lewis John Ellery (Government Astronomer, Vic) and Charles Todd (Superintendent of Telegraphs, SA) organise the work required.

By 1867 it was agreed that :

(1) the difference in longitude between the Observatories at Sydney and Melbourne be made by means of the electric telegraph, to eliminate the existing discrepancy;

(2) the longitude of the Sydney Observatory be the datum from which the boundary is measured, (this datum longitude being derived from the arithmetic mean of Stone’s longitude derived from Scott’s 1859 & 1860 work [10h 4m 47.32s] *and the longitude calculated by applying the difference Sydney-Melbourne* [from (1) above] *to the longitude of Melbourne Observatory 9h 39m 54.8s;*

(3) an observing station on the Murray should be established with telegraphic communication with Sydney and Melbourne and that its longitude should be determined by voltaic signals exchanged with Sydney on two or more clear nights. The signals to be transits over the meridian, or rather, over the several wires of the telescope, of certain stars previously selected, observed at both places and recorded on the Sydney chronograph [accurate clock with paper tape recording facility]. *Similar operations to be made in respect to the Melbourne Observatory;*

(4) that on the determination of the latitude and longitude of the observing station, the calculated distance to the boundary line [141st meridian] be set off.

In May 1868, Todd, accompanied by Arthur Bevan Cooper, Deputy Surveyor General, South Australia, occupied the observing station near the boundary (at Lake Littra, South Australia), Smalley occupied the Sydney Observatory, and Chief Assistant White occupied the Melbourne Observatory. At that time the *electric* telegraph between Adelaide and Sydney ran north of the Murray River to Wentworth and then east to Sydney. Installing a temporary spur telegraph line to connect the Lake Littra site to the main line would therefore not have been onerous.

On 9 and 10 May the transits of ten stars, over the meridian of the Observatories at Sydney and at the boundary were recorded simultaneously at both stations, and on 13 and 14 May the transits of 21 stars were similarly recorded at the boundary and at the Melbourne Observatory. Time signals were also exchanged between the Observatories at Melbourne and Sydney. After Smalley telegraphed his acceptance of the determination, the line to the calculated position of the boundary was laid off and the 141st meridian was then carefully run south nearly two miles. In November, Smalley met Todd at Wentworth and formally accepted, on behalf of New South Wales, the boundary as marked on the ground. Its position was 2 miles 44 chains 68.2 links (4.1 kilometres) due east of the Lake Littra observatory meaning Victoria was occupying a strip of South Australia some 2¼ miles wide.

The results from the May 1868 telegraphic exchanges gave :

Difference in longitude	**Hours**	**Minutes**	**Seconds**
Boundary-Sydney	0	40	59.72
Boundary-Melbourne	0	16	03.77
Melbourne –Sydney	0	24	55.81

These results led to the adoption of a value for the datum longitude of the Sydney Observatory as prescribed in point *(2)* above as 10h 4m 48.97s (151º 12’ 14.5”).



Figure 8 : The cairn erected in 1868 near Lake Littra, SA, after Charles Todd determined the position of the 141st Meridian of Longitude

South Australia then laid claim to the 2½ mile strip occupied by Victoria and the dispute continued until February 1911. The case was taken before the High Court of Australia which decided that as the marked boundary had been accepted as such by the two States concerned, the fact that it did not exactly coincide with the 141st meridian of east longitude did not warrant the Court ordering the re-adjustment claimed by South Australia.



Figure 9 : Original Cable Station Broome, WA

When the submarine telegraphic cable was brought ashore at Darwin in late 1871 it connected Australia to the rest of the world via Java (Banjoewangi - today Banyuwangi) and Singapore. But it was the completion of the Overland Telegraph (the Port Augusta to Port Darwin telegraph line) on 22 August 1872 that really ended Australia's isolation from the northern hemisphere. By 1877 all capital cities except Perth were telegraphically interconnected. In 1889, a link from Batavia (Jakarta) was bought ashore at Broome then went overland to Perth. A further submarine telegraphic cable via the Cocos-Keeling Islands made landfall in Perth in 1901. This cable was part of the so called *Red Route* as it traversed only British controlled

countries. The global circuit was completed in 1902 with a submarine telegraph cable across the Pacific between Canada and Australia via Fiji and Norfolk Island. This cable was bought ashore at Southport south of Brisbane.

The official survey of the 29ºS parallel, the border between New South Wales and Queensland was conducted by John Cameron (NSW) and George Watson (Qld) between 1879 and 1881. Astronomical observations were taken at the Barringun telegraph station to determine latitude and longitude. Following these observations, the *zero obelisk* was erected on the banks of the Warrego River just north of the town. From this mark, the border to the west was marked first followed by the section to the east. It appears that while the Barringun telegraph station was a temporary observatory its longitude was never determined or checked by an exchange of clock signals.



Figure 10 : The Zero Obelisk north of Barringun, Queensland

There was also a proposal in 1877, by Frederick Drake-Brockman of the Western Australian Survey Department, to fix the 129th meridian which was the border between Western Australia, the Northern Territory of South Australia and South Australia itself. His plan, which was really only for the northern extent of the border, firstly entailed fixing the longitude of the Halls Creek telegraph station by an exchange of clock signals. Longitude would then be carried by a survey connection to an existing survey network which ran toward the border. From the nearest point to the border on that existing survey, a new survey would then establish the actual position of the meridian. The meridian would then be marked northwards to the coast. History shows his proposal never eventuated.

An arrangement was made between the British and Australian authorities in 1882 to complete the longitude chain Greenwich-Australia by the telegraphic method. Captain L. Darwin was sent to Singapore, Pietro Baracchi, Government Astronomer of Victoria, travelled to Port Darwin (now Darwin) and Captain Helb of the Netherlands India Staff of Batavia to Banjoewangi (Java). Their aim was to determine the time intervals Banjoewangie-Singapore and Banjoewangie-Port Darwin as well as the direct interval Singapore-Port Darwin.

Clock signals were exchanged by Port Darwin with Banjoewangie on 4 nights, with Singapore on 8 nights, with Adelaide on 6 nights, and with Melbourne on 4 nights, between the dates 28 January and 2 March 1883, inclusive.

These operations resulted in the meridian of Port Darwin then becoming the origin of Australian longitudes and the following results were obtained and adopted at the time :

1883 Longitude	**Hours**	**Minutes**	**seconds**	**degrees**	**minutes**	**seconds**
Port Darwin	08	43	22.49	130	50	37.4
Adelaide Observatory	09	14	20.30	138	35	04.5
Melbourne Observatory	09	39	54.14	144	58	32.1
Sydney Observatory	10	04	49.54	151	12	23.1
Hobart	09	49	19.80	147	19	57.0

During the 1870s, observations for longitude consisting of both transits of the moon and occultations of fixed stars by the moon continued. Dr Georg Friedrich Julius Arthur von Auwers of Berlin used all these observations to derive a longitude for the Sydney Observatory of 10h 4m 49.60s (151º 12’ 24.0”). The comparison of results from the entire chain of telegraphic longitudes completed in 1883 with those derived by Dr Auwers from lunar observations showed near perfect agreement.

During the next twenty years the longitude of the following places was established by the telegraphic exchange of time signals :

Brisbane Observatory in the years 1884, 1891, and 1892;

Broome, Fremantle, and Albany in 1890 and 1891; and

Perth Observatory in 1899 and 1901.

The origin for these longitudes was still the 1883 (Port of) Darwin meridian.

The 138th meridian defines the border between Queensland and the Northern Territory. When this border was established, however, the Northern Territory was still part of South Australia and later took its name from being called *the Northern Territory of South Australia*. The marking of this border was completed in September 1886 with the work carried out by South Australian Government Surveyor Agustus Poeppel who was later replaced by John Carruthers, due to Poeppel's severe eye problems.

Queensland could not provide a survey team for the actual work but agreed to pay half the cost but before doing so wanted to independently check the South Australian work. The check comprised a survey traverse from the Boulia telegraph station to the border and the subsequent fixing of Boulia's longitude by telegraph from the Brisbane Observatory. At that time Boulia was the closest town to the border with an existing telegraph connection to Brisbane.

Staff surveyor Cecil Twisden Bedford was chosen by the Queensland Surveyor General William Alcock Tully to carry out the survey. Bedford traversed westwards from Boulia in late 1885 and connected to the border between the 255 mile and 255 mile 77 chain posts in February 1886. A 2014 study of Bedford's plans of survey showed he traversed 147 miles (Kitson, 2014). The 135 miles quoted for Bedford's traverse in some documents is probably just an estimate as Bedford's survey was somewhat governed by the terrain so did not take a direct route from Boulia to the border.

An exchange of time signals via the electric telegraph between the Brisbane Observatory and the telegraph office at Boulia was undertaken by Queensland staff surveyors Robert Hoggan and Robert Grant McDowall in 1887. The check by Queensland concluded that the South Australian work was acceptable.

As mentioned above, the global telegraphic circuit was completed in 1902 with a submarine telegraph cable across the Pacific Ocean between Canada and Australia. Canadian Astronomers were keen to "*close a complete longitude circuit round the earth* and by 1903 Otto Klotz of the Dominion Observatory Ottawa, Canada had concluded the necessary work. **"***A second fundamental meridian* [through Southport, south of Brisbane] *was thus established in 1903 on the eastern coast of Australia, the longitude of which was based entirely on the telegraphic method, and was quite independent of any other Australian longitude previously determined*" (Commonwealth Government Handbook, 1914).

Following the measurement of the arc Southport-Sydney, by exchange of clock signals, the entire circuit round the earth was closed. A comparison of the two independent values for the Sydney Observatory was then available. It should be noted that other work had also been carried out by this time thus improving the longitude values for Australian meridians.

The values adopted by Klotz gave the following longitudes for the Sydney Observatory :

Longitude	**Hours**	**minutes**	**Seconds**	**Degrees**	**Minutes**	**seconds**
Via Asia	10	4	49.355	151	12	20.33
Via the Pacific	10	4	49.287	151	12	19.31
Closing error	0	0	0.068	0	0	1.02

This misclose represented a distance of just over 30 metres on the ground.

The most probable longitude values for those Australian stations which were part of the *Canadian circuit* were given as :

Longitude	**Hours**	**Minutes**	**seconds**	**degrees**	**minutes**	**seconds**
Port Darwin	08	43	22.28	130	50	34.20
Melbourne Observatory	09	39	53.93	144	58	28.95
Sydney Observatory	10	04	49.33	151	12	19.95
Southport	10	13	39.82	153	24	57.30

(To see how close these 1903 values were to those adopted later, note that in the 1950s and 60s Australia published the R502 series of maps. This first complete map coverage was at 1:250,000 scale. In south-east Australia the horizontal datum was based on the then latitude and longitude of the Sydney Observatory quoted as 33º 51' 41.10"S, 151º 12' 17.85"E (10h 4m 49.19sE). Thus some fifty years after Klotz's probable vale for the longitude of the Sydney Observatory was given the value had been refined by 0.14 seconds of time amounting to some 60 metres on the ground).

Despite the close agreement in longitudes obtained by Klotz in 1903, the result was considered to be *more good luck than good management*. This view was based on the fact that on the circuit via Asia there was up to ten individual major segments of telegraph that had had their difference in longitude measured by different methods, operators, equipment, line signal quality etc and some segments needed re-observing to ensure standardisation. *Wire* was, however, about to be replaced by *wireless*!

Australian longitudes by "wireless"

Guglielmo Marconi made the first wireless transmission in 1897. By 1907 Canada was using this technology to broadcast an automated daily time signal to ships at sea from its Marconi station at Halifax. The French *Bureau des Longitudes*, in conjunction with the French military, began using the Eiffel Tower to also transmit time signals twice daily. These signals were primarily intended to enable ships' clocks to be updated more regularly and not just while in port, resulting in more reliable longitude determinations. Nevertheless, anyone with the requisite receiving equipment could receive these time signals when in range.

The value of these *broadcast* time signals was quickly realised and more transmitters worldwide were established. As early as 1912 the *Bureau des Longitudes* hosted the *Conference Internationale de l'Heure* (*International Time Conference)* at which it was agreed that from 1 July 1913 all stations transmitting time signals would send out the same automated signals conforming to a set sequence.

In 1920, the Western Australian Government Astronomer, Harold Burnham Curlewis, raised the possibility of using wireless time signals to determine the position of the border between South Australia/Northern Territory and Western Australia as close to the 129th meridian as possible.

In November 1920, a party comprising George Dodwell (Government Astronomer, South Australia), Clarence Maddern (Government Astronomer Assistant, South Australia), Harold Curlewis (Government Astronomer, Western Australia), Clive Hambidge and J. Crabb (South Australian Survey Department) and Warrant Officer Victor Bowen (in charge of the wireless apparatus lent by the Defence Department) proceeded to Deakin, Western Australia on the Trans-Australian Railway.



Figure 11 : The Deakin Pillar, SA (left), and the Kimberley Obelisk, WA (right), with (L-R) P.M. Durack, M.P. Durack and Sir William Campion

To permit simultaneous, astronomical observations for longitude by three different methods (meridian transits, equal altitudes and almucantar transits) three observing pillars were constructed. One pillar was later named the *Deakin Pillar*. This survey generated world-wide scientific interest and involved the co-operation of the Astronomer Royal and the Royal Observatory, Greenwich. Wireless time signals were transmitted by the French Wireless Service from the *Observatoire de Lyon*, France, between 17 and 24 November 1920. Wireless time signals were also sent from the Adelaide Observatory transmitted by the Adelaide Radio Station to enable the times of the observations, as read from the chronometer, to be adjusted for any error.

With everything in place and tested, the actual observing and determination of longitude took place in April-May 1921. The survey party comprised Dodwell, Maddern, Curlewis, Hambidge as well as Arthur Williams and HS Duncan (South Australian Survey Department). To attempt to ensure best possible time, wireless time signals from Lyons, Bordeaux (France), Annapolis (USA), Adelaide and Perth were received at Deakin. This was the first longitude determined from *wireless* time signals.

Based on the longitude determined by the 1921 observations, the *Deakin Obelisk* was constructed 2.82 kilometres east of what was later named the *Deakin Pillar*. A line due north and south through the *Deakin Obelisk* indicated the border.

Messrs Dodwell, Curlewis, Hambidge and Maddern then left for the Kimberley Region on the *SS Bambra* on the 28 May 1921. The group was then joined by Michael Durack (Member of Parliament for the Kimberly Region 1917-1924). Durack took them to a point 29 kilometres north of the homestead on his Argyle Downs pastoral property which he perceived was close to the 129th meridian.

Two pillars (the most easterly known as the *Austral Pillar*) were established allowing Transit and Almucantar instruments to be used for simultaneous astronomical observations. Accurate time was once again available by the reception of wireless time signals from Bordeaux, Lyons, Annapolis, Adelaide and Perth. The results showed that the observation pillars were approximately 2.4 kilometres east of the border.



Figure 12 : The observing site (left) and wireless reception aerial (right) near the 129th Meridian of Longitude, Kimberley Region, WA

The boundary agreement (Western Australian Crown Law Deeds No. 9023) was signed on 4 November 1922. Prime Minister William Morris Hughes signed for the Commonwealth and Acting South Australian Premier John Bice and Western Australian Premier Sir James Mitchell signed for their respective state governments. This agreement essentially defined the border to be a line due north and south through a point (the *Kimberley Obelisk* which was not established until 1927) to be determined from the astronomical stations near Argyle Downs Homestead and the point (the *Deakin Obelisk*) on the boundary in the vicinity of Deakin. Importantly, the physical marking would indicate the boundary despite any future (more accurate) observation!

The *Kimberley Obelisk* was erected in 1927. Its position indicated the 129th meridian and was established by a survey by the Western Australian Department of Lands and Surveys that had the *Austral Pillar* as its origin.

At the 1967 meeting of the National Mapping Council (NMC), which consisted of the State's Surveyors General, the then chairman and Director of National Mapping, Bruce Lambert, was of the opinion that the 1922 agreement should be amended so that the Western Australia border was a straight line. However, as Western Australia had already marked and adopted some 300 kilometres of the border any such change would cause them difficulty. The three states ageed that the spirit of the 1922 agreement should be maintained and the Council unanimously resolved that there was no justification in seeking a variation to the 1922 agreement. Consequently, it was determined that two marks were to be placed on the 26° South Latitude; one due north of the *Deakin Pillar*, the other due south of the *Austral Pillar*. On the suggestion of Bruce Lambert the *step* in the border was named *Surveyor Generals Corner*. On 4 June 1968 two concrete pillars 126.958 metres (AGD66) apart, were completed to mark *Surveyor-Generals Corner*.

As far back as 1891 the then Surveyor General for Queensland, Archibald McDowall saw the need to accurately fix the position of the state’s isolated outback towns. As any existing Queensland survey network would not encompass this outback region for the foreseeable future, McDowall promoted the use of astronomical determination of position as a local basis for any mapping.

Despite the unanimous agreement for a Commonwealth national geodetic survey at the Conference of the Director of Commonwealth Lands and Surveys, the Surveyor General and the Government Astronomer of New Zealand, and the Surveyors General of the States of the Commonwealth of Australia of May 1912, nothing immediately eventuated.

Thus is 1927, the first use of broadcast wireless time signals allowed Queensland surveyors to obtain accurate position from astronomical observations. Station 4QG transmitted the time signals using a radio tower adjoining the Survey Office Observatory in Brisbane, at the corner of George and Elizabeth Streets.

Longitude anywhere

As with any technology developments led to more powerful transmitters being used to broadcast time signals over a greater range, and wireless receivers became less specialised and more transportable. Thus it was that anywhere a wireless receiver could be transported, accurate time could be obtained. The requirements to establish longitude suitable for marine navigation were in place but on land a greater degree of accuracy was a necessity as land ownership now needed to be uniquely described.

The almost complete lack of accurate mapping for the Australian continent became starkly apparent during World War II. The wartime need was met by the instigation of the Emergency Mapping Scheme in November 1940. As the end of the war approached national development became the focus and to enable such development to proceed, maps were a basic requirement. The National Mapping (Nat Map) Section within the Property and Survey Branch of the Department of Interior was established in 1947 but with an accommodation shortage in Canberra, in 1948 the Photogrammetric Survey element was accommodated in Melbourne. In that same year the first Nat Map Astrofix for a continental (the R502 series) standard, map coverage was taken.

An Astrofix was simply an observation of stars from which could be computed the latitude and longitude of the observer. Apart from receiving a broadcast time signal an Astrofix was completely self-contained. This made them the ideal observation for basic map control and some 2,540 Astrofixes were used in the production of the continental R502, 1:250,000 scale map series completed in 1968. The accuracy of these observations was estimated to be ±3 seconds of arc (±0.2 seconds of time) or about ±75 metres. Considering that until this era position finding relied on the *lunar distance method* with an accuracy in the order of several thousand metres, Astrofixes were relatively precise.



Figure 13 : Map showing the locations of the 2,540 Astrofixes that were used in the production of the continental R502, 1:250,000 scale map series

Also, by 1940 it had been shown that the quartz oscillator was more accurate than the best existing mechanical clocks, then mainly used as a time standard in astronomical observatories. Worldwide, time standard laboratories began switching from mechanical to quartz-based clocks.

This progress had come about through the telephone replacing the telegraph and from broadcast radio evolving. The problems of maintaining a stable electrical frequency and then its monitoring had emerged. During his investigations Canadian, Warren Alvin Marrison developed a highly accurate clock in 1927, using the regular vibrations of a quartz crystal in an electrical circuit. Quartz clock operation was based on the piezoelectric (electricity resulting from pressure) property of quartz crystals. The interaction between mechanical stress and electric field of a quartz crystal in a suitable electronic circuit causes the crystal to vibrate (of the order of 10,000,000 pulses per second) and generate a constant frequency electric signal. When this signal was divided up electronically until the number of pulses reaches 60 per minute the output was fed to the (analogue and later digital) clock display.

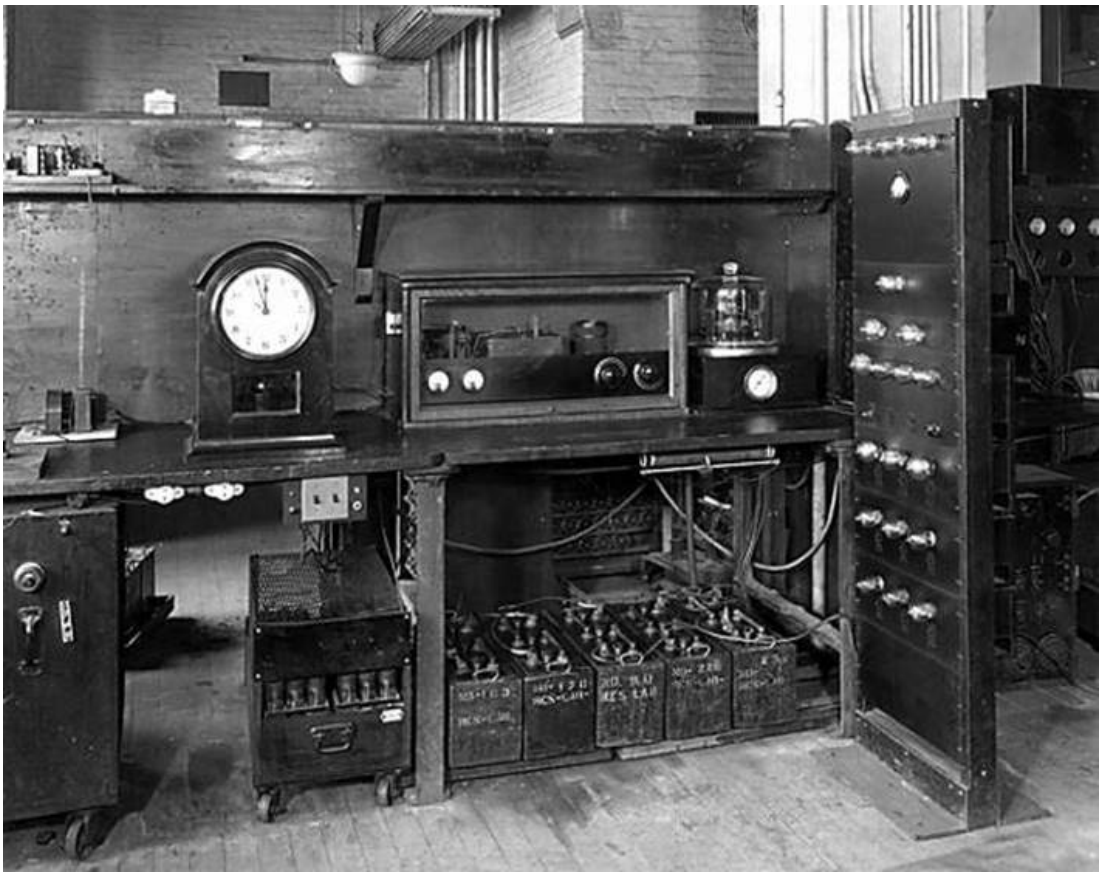


Figure 14 : Warren Alvin Marrison's quartz clock of 1927

Accurate time via the quartz crystal was an integral part of the first electronic distance measuring (EDM) equipment used in Australia. Dr Erik Osten Bergstrand of Sweden invented a device to accurately measure the speed of light. In 1947 Bergstrand took his instrument to a 7,734 metre baseline and obtained a measurement of 299,793.1 ±0.2 kilometres per second (today 299,792.458 kilometres per second) for the speed of light. Having this value Bergstrand could adapt his device to measure distance. In conjunction with AGA (Aktiebolaget Gasaccumulator) of Sweden the Geodimeter (GEOdeticDIstanceMETER) model NASM-1 appeared on the market in 1953. In May 1954 National Mapping's NASM-1 (one of only ten of this model produced) arrived in Australia. The EDM experience gained by National Mapping, via the Geodimeter, saw EDM then significantly impact the Geodetic Survey of Australia in future years.

While highly reliable and inexpensive no two quartz crystals emit the same frequency. In search of a better *standard* the then American National Bureau of Standards announced in 1950 that it had developed the world's first *atomic* clock; its source - the vibrations of the ammonia molecule. This technology developed rapidly such that in 1967 the 13th General Conference on Weights and Measures defined the second on the basis of vibrations of the caesium atom; the world's timekeeping system no longer had an astronomical basis. Finally by 1999 the American National Institute of Standards and Technology (NIST)-F1 had begun operation with an accuracy to about one second in 20 million years, making it at that time the most accurate clock ever made (a distinction shared with a similar standard in Paris).

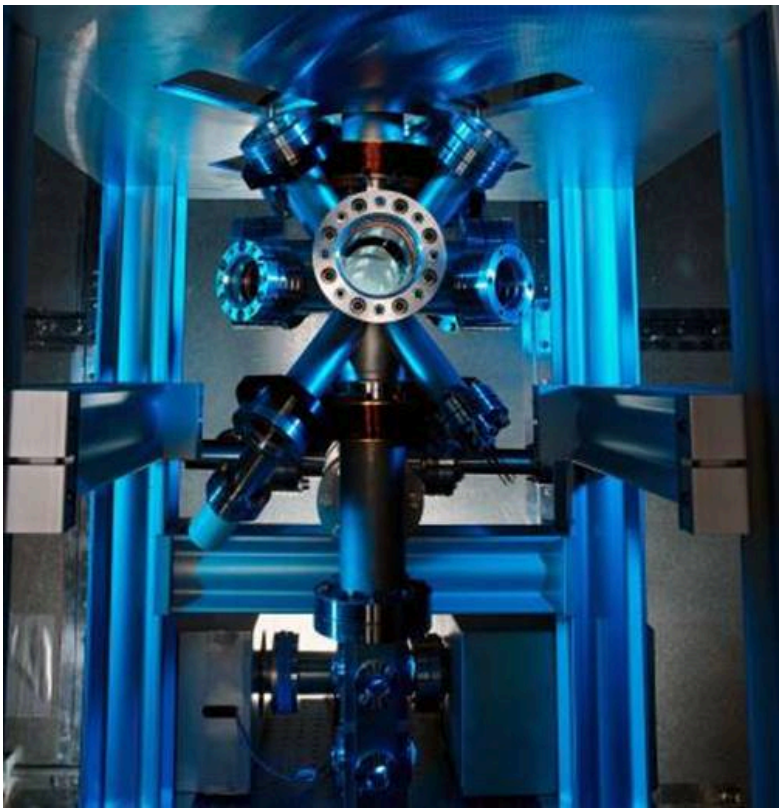


Figure 15 : National Institute of Standards and Technology (NIST)-F1

Similarly the accuracy and quality of the Astrofix observing equipment was maximised. Combined with the observation of more stars over at least two nights and automating time reception by using a *chronograph* later Astrofixes were found to provide an accuracy of better than 1 second of arc (0.06 seconds of time) or around 25 metres. On land in Australia, the determination of longitude was one step away from its apex!

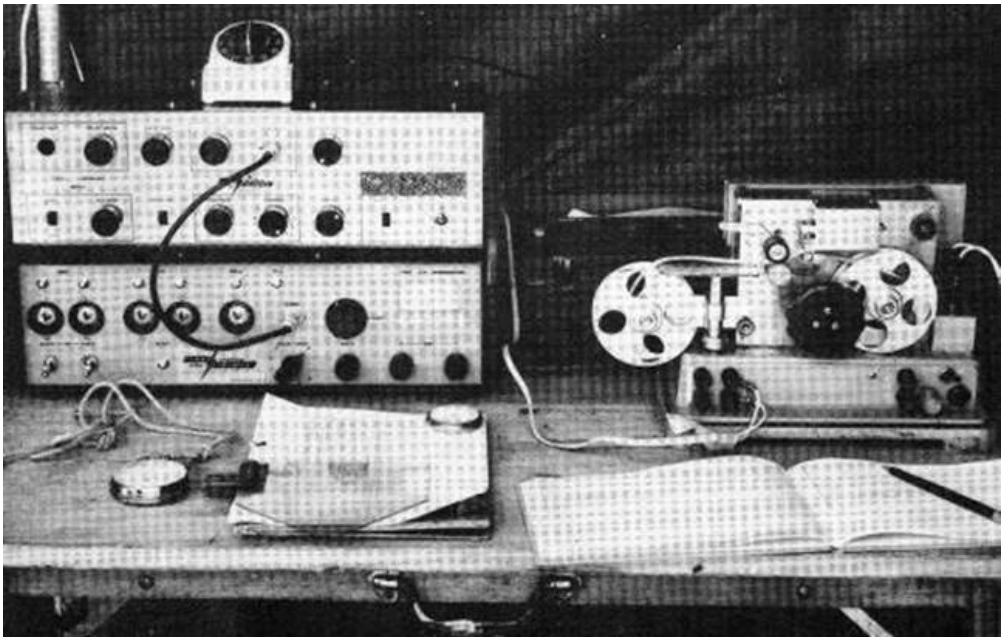


Figure 16 : Chronograph with wireless time-signal receiver and paper tape recorder used by National Mapping for Astrofixes, circa late 1960s

Longitude today

Accurate time is the basis of Global Positioning Systems (GPS) and *GPS chips* are in phones and cameras as well as hand-held and automotive navigation units. With a clear view of the sky, longitude may be obtained to better than 10 metres and down to the millimetre with some specialised equipment.



Figure 17 : GPS *Chip*

After some 225 years, time and hence longitude is now available anywhere (provided the GPS batteries are charged) and naturally the question arises "*just how good were the techniques of longitude by wire and wireless"?*

The Australian experience shows these techniques provided longitude to 1 kilometre or better which perhaps at first sounds a little disappointing. When it is considered, however, that this accuracy was achieved when the longitude of the reference meridian was still somewhat uncertain and is now compared with a global, satellite based context of fixing longitude and add in the need to negotiate some of Australia's still most inhospitable country then this level of accuracy is no less than astounding!

Prepared by Paul Wise June - December 2014

Addendum 2017

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Annexure A

This article was found in *The Surveyor*, of January 7 1895, Vol. VIII, No. 1, p. 4, and has been reformatted for easier reading.

STANDARD TIME

ADMINISTRATION OF THE NEW ACT. UNIFORM TIME THROUGHOUT QUEENSLAND. THE TIME-BALL IN BRISBANE.

The following particulars have been supplied by the Surveyor-General :-

The provisions of the *Standard Time Act*, which recently passed through both Houses of our Legislature, will come into operation on 1st January, 1895. Thereafter, one time will be kept throughout the colony, uniform with the mean time of the 150th meridian east of Greenwich, or exactly ten hours earlier than Greenwich time.

Simultaneously with this improvement in the method of reckoning time, it has been decided to officially initiate the working of the time-ball. As most of our readers will have observed, this was erected a few months on the Signal Tower, Wickham Terrace, and has been working very satisfactorily ever since, but no announcement of the fact has been made, pending the approval by Parliament of the *Standard Time Bill*.

The method of making the signal will be similar to that adopted at Greenwich and other places where such apparatus is in use, and will be as follows : The ball will be hoisted half-way up the mast at 12.55 p.m., and close up at 12.57 p.m. It will be dropped automatically by electric signal from the standard clock at the Survey Office Observatory at 1 o'clock, mean time of the 150th meridian east of Greenwich, which is later than the mean time of the Survey Office Observatory by 12min. 06.40sec. For rating chronometers the time should be noted at the instant the ball begins to descend. If from accident the drop fail in accuracy, the ball will be immediately hoisted half-mast, or if it fails to drop

it will be gradually lowered to half-mast, where it will remain till 1.57 p.m., when it will be hoisted close up again, and dropped at 2 p.m.

The signals will not be made on Sundays or on public holidays. On the 1st January, however, it will be dropped at every hour from 6 a.m. till 1 p.m., in order to give the public an opportunity of correcting their clocks and watches to the new standard of time, and, as an additional aid to this object, it is proposed to fire the gun at midnight and 1 p.m. on this day. It is anticipated that the residents of the locality will not object to the noise on these occasions, in consideration of the advantage which it will otherwise be.

The ball was made and erected under the supervision of Capt. Almond, and the electro magnet detent and fittings were designed and constructed by Mr. S.H. Smith, electrician, of the mechanical branch of the Post and Telegraph Department. The ball is held in position, when hauled up, by placing a stirrup, attached to the rope, under a niche in the detent. When the clock makes connection on the line the armature of the detent is attracted, the stirrup is released, and the ball descends. The error of the clock is determined daily, either by astronomical observations or from the rates of standard chronometers, and is eliminated at 12.30 p.m. by means of a special electro-magnetic attachment, without stopping or touching the clock. This consists essentially of a pair of electromagnets secured to the back of the case, so that a permanent magnet affixed to the pendulum just clears them when it swings. By means of a commutator a current can be sent through the coils of the electro-magnets, so as to effect an attraction or repulsion of the permanent magnet on the pendulum, thus producing a retardation or acceleration of its vibrations as may be required. In this manner, with the aid of three Daniell's cells, an error of half a second may be eliminated in eight minutes. The control and proper working of the apparatus is in the hands of Mr. T.D. Fraser, of the Surveyor-General's Department. -- *Brisbane Courier*, Dec. 6, 1894.